State vs Cache:
Understanding Long-Horizon Evolving State

Concept Summary — DataForge 2026 / Pathway Track

CORE CLAIM

A fixed-shape recurrent state can process a sequence of unbounded duration without allocating a new memory slot for every token, but information stored in that fixed state can interfere, causing retrieval failures. This claim is demonstrated in a small interactive toy retrieval experiment. It is not a benchmark result and does not reproduce a trained BDH model, its training procedure, or published benchmark results.

THE QUESTION

What happens when more key-value facts must be represented inside a fixed-size recurrent state, and how does increasing the state dimension affect retrieval interference?

THE EXPERIMENT

The experiment generates the same stream of random key-to-value facts for two memory systems. **1. Growing Cache:** A JavaScript Map stores one entry per fact, yielding 100% exact key lookup accuracy. Memory grows linearly; facts remain isolated. **2. Fixed Recurrent State:** A D×D matrix `sigma` is initialized to zero. Each key/value pair is written using an additive outer product:

```
sigma_t = sigma_(t-1) + x_t^T v_t
```

For a query, the system computes `o_t = x_t · sigma_t` and selects the predicted value by nearest cosine similarity. All facts are superimposed into the same fixed matrix. Because random keys are not perfectly orthogonal, one key can activate components associated with other stored facts. As the information load increases, this cross-talk can cause incorrect nearest-neighbor retrieval. D represents available representational capacity, not a guaranteed orthogonality bound.

INTERACTIVE VARIABLES

The learner controls the number of stored facts (N) and state dimensionality (D). The browser recomputes the experiment live when either control changes. The interface exposes cache/state retrieval accuracy, a sigma heatmap, individual query results, ground truth, and a Query Inspector, making the claim experimentally inspectable.

SENSITIVITY CHECK

25-seed sensitivity check — not a formal statistical study
Production configuration: N=18, D=8 (Production seed 0xDEADBEEF yields 50.0% state accuracy, 100% cache accuracy).
Across 25 seeds at N=18, D=8: Mean: 42.7% | Median: 44.4% | Range: 22.2% to 66.7% | Within 35% to 65%: 18/25 seeds.

The production 50.0% result is reasonably representative of the sampled behavior. It is close to the sampled mean and median, and 18 of 25 seeds fall within ±15 percentage points of it. Individual random seeds vary because different random key vectors have different degrees of overlap.

CONNECTION TO DRAGON HATCHLING (BDH)

The toy's write/read mechanism shares the mathematical structure of the idealized formulation described in Pathway's "*From attention to synapses: deriving BDH*" (Chapter 2 of the BDH Explainer Series). This is a **mechanism-level analogy** and a toy implementation of one isolated mathematical mechanism. It is NOT a reproduction of the trained BDH architecture. It uses an engineered key/value retrieval task and a dense D×D state, whereas real BDH includes additional learned dynamics and nonlinearities.

WHAT THE TOY OMITS

- ReLU nonlinearity and enforced sparse, non-negative activity
- Transition operator U for positional/decay effects
- Low-rank factorization (E, Dx, Dy) used in the practical BDH formulation
- Trained parameters, training procedure, and published benchmark evaluations

KEY TAKEAWAY

A growing cache preserves exact recall by allocating additional memory for each fact. A fixed recurrent state keeps its memory footprint bounded as the sequence grows, but multiple facts must share the same representational structure. The resulting interference provides an observable example of the trade-off between bounded state size and information capacity.

EVIDENCE / REFERENCES

- 1. Pathway, "The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain" (arXiv 2509.26507, 2025).
- 2. Pathway, "Reasoning at a Fraction of the Compute" and the accompanying BDH-CQ technical report (Aug 2026).
- 3. "A Hippocampus for Linear Attention: An Exact Memory for What the Recurrent State Forgets" (arXiv 2607.02303, 2026).
- 4. "LoLA: Low-Rank Linear Attention With Sparse Caching" (arXiv 2505.23666, 2025).